

EXECUTIVE SUMMARY

Factory-to-Customer Shipping Route Efficiency Analysis *Nassau Candy Distributor*

Prepared by: Anushka Singh

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1. Project Overview

This project analyzes shipment data for Nassau Candy Distributor to understand factory-to-customer shipping route efficiency. The analysis focuses on shipping lead time, route-level performance, shipment volume, delay frequency, geographic patterns, and differences across shipping modes. The objective is to identify routes and geographic areas that may require closer operational monitoring and to provide data-driven insights that can support logistics planning and performance monitoring.

2. Data & Methodology

The analysis was performed using 10,194 shipment records containing information on orders, shipping dates, shipping modes, customer locations, products, sales, costs, and gross profit. The dataset covered orders placed between January 2, 2024 and December 31, 2025, with supplied shipping dates ranging from June 30, 2026 to June 28, 2030.

The data was first checked for missing values, duplicate records, invalid dates, and inconsistencies in geographic and shipping-mode fields. Shipping Lead Time was calculated as the difference between Ship Date and Order Date. Each product was mapped to its corresponding factory using the project-provided product-to-factory mapping.

For route-level analysis, a route was defined as Factory → Customer State/Province. Route performance was evaluated using shipment volume, average lead time, lead-time variability, delay frequency, and a normalized Route Efficiency Score. A delay threshold of 1,274 days, equal to the dataset median lead time, was used as an analytical reference rather than an official delivery-service standard.

Geographic and shipping-mode analyses were also performed to identify differences in shipping performance and areas that may require further operational investigation.

3. Key Findings

The analysis identified substantial variation in shipping performance across factory-to-customer routes. A total of 196 unique routes were identified from the 10,194 shipment records.

Route shipment volumes were concentrated in several major destinations. The highest-volume routes included Lot's O' Nuts → California (1,125 shipments), Wicked Choccy's →

California (823 shipments), Lot's O' Nuts → New York (645 shipments), and Lot's O' Nuts → Texas (569 shipments).

Average shipping lead time also varied considerably between routes. Some routes had average lead times of approximately 906–909 days, while several other routes had average lead times above 1,637 days. This variation highlights the importance of evaluating shipping performance at the route level rather than relying only on overall averages.

Using the project-specific analytical threshold of 1,274 days, 4,302 shipments (42.20%) were classified as delayed. This threshold is based on the dataset median and is not an official service-level agreement.

Geographic analysis showed that Washington and Tennessee were notable areas for further monitoring because they combined meaningful shipment volumes with relatively high average lead times and delay frequencies. Washington recorded 506 shipments, an average lead time of 1,360.66 days, and a delay frequency of 52.77%, while Tennessee recorded 183 shipments, an average lead time of 1,391.49 days, and a delay frequency of 55.19%.

The results also showed differences between shipping modes. Standard Class accounted for the largest shipment volume, with 6,120 shipments and an average lead time of 1,314.33 days. The observed differences between shipping modes are descriptive and should not be interpreted as evidence that a particular shipping mode directly causes shorter or longer delivery times.

4. Recommendations

Based on the observed route, geographic, and shipping-mode patterns, the following actions are recommended:

1. Prioritize high-volume routes for monitoring.

Routes serving major shipment destinations such as California, New York, Texas, and Washington should be monitored regularly because changes in their performance can affect a significant number of shipments.

2. Monitor potential geographic bottlenecks.

Areas such as Washington and Tennessee should receive further investigation because they combine meaningful shipment volumes with relatively high observed lead times and delay frequencies.

3. Establish route-level performance monitoring.

Regular monitoring of average lead time, shipment volume, delay frequency, variability, and route efficiency can help identify changes in route performance over time.

4. Investigate routes with high variability.

Routes showing substantial variation in shipping lead time should be examined further to understand the factors contributing to inconsistent performance.

5. Compare shipping modes within individual routes.

Shipping-mode performance should be evaluated in the context of specific routes rather than relying only on overall averages.

6. Combine efficiency with shipment volume.

Route efficiency scores should be interpreted together with shipment volume and delay frequency so that low-volume routes do not receive disproportionate attention.

7. Improve logistics data availability.

Additional information such as actual delivery dates, promised delivery dates, carrier details, transportation distance, transportation cost, shipment weight, and reasons for delay would support more detailed logistics analysis.

5. Conclusion

The analysis provides a route-level view of shipping performance across the Nassau Candy Distributor dataset. Using 10,194 shipment records and 196 factory-to-customer routes, the study examined shipping lead time, shipment volume, delay frequency, route variability, geographic patterns, and shipping-mode differences.

The results show that shipping performance varies considerably across routes and that shipment volume should be considered alongside lead-time and delay measures when identifying areas for operational monitoring. Geographic analysis also highlighted areas such as Washington and Tennessee for further investigation, while differences across shipping modes provided additional context for understanding observed shipment performance.

The analysis demonstrates how structured data analysis and visualization can transform shipment records into actionable logistics insights. The proposed route-level monitoring approach can provide a foundation for continued performance tracking and more detailed investigation as additional logistics data becomes available.

6. Limitations

The analysis has several limitations that should be considered when interpreting the findings. The supplied dataset contains an unusual distribution of order and shipping dates, resulting in relatively large calculated shipping lead times. These values were retained because they are present systematically in the supplied data and there was no verified basis for modifying them.

The 1,274-day delay threshold is an analytical reference based on the dataset median and is not an official delivery-service standard. Some routes also have relatively small shipment volumes, so their performance measures should be interpreted cautiously.

The dataset does not contain several logistics variables that could explain the observed differences, such as actual delivery dates, promised delivery dates, carrier information, transportation distance, shipment weight, and specific reasons for delays. Therefore, the analysis identifies patterns and areas for further investigation rather than establishing causal explanations.